University of Cape Town

Supplementary Examinations – January/February 2004
Department of Mathematics and Applied Mathematics

MAM200W

Module 2LA : Linear Algebra

Time : $2\frac{1}{2}$ hours **Full Marks: 75**

Marks available: 78

Your answers must be clearly and carefully presented. Sloppily presented solutions will be penalized. You may use approved calculators.

This paper consists of 3 pages.

1. Let S be the subset of P_3 defined by

$$S = \{x^3 - 3x, x^2 + 2, 2x^2 - x, x + 4\}.$$

Answer the following questions, giving reasons for your answers.

(a) Find a basis B for span(S).

(b) What is dim(span(S))?

(c) Is $x + 1 \in \text{span}S$?

(d) Is there a $p(x) \in P_3$ such that $B \cup \{p(x)\}$ is a basis for P_3?

(e) Is the set $V = \{p(x) \in P_3 : p(-1) = p(1)\}$ a subspace of P_3?

(f) For $p(x), q(x) \in P_2$, put $\langle p(x), q(x)\rangle = \int_{-1}^{1} p(x)q(x)\,dx$. You may assume that this defines an inner product on P_2. Let $p(x) = x - 2, q(x) = 2x^2$. Find $||p(x)||$. Is $p(x)$ orthogonal to $q(x)$?

[17]

2. Let the function $T : M_{2\times2} \to M_{2\times2}$ be defined by $T(A) = \frac{1}{2}(A - A^T)$.

(a) Show that T is a linear transformation.

(b) Find dim(ker(T). Show how you arrived at your answer.

(c) Find a basis for ran(T).

(d) Is T one-to-one? Is it onto? Give reasons for your answers.

(e) Find the matrix representing T with respect to the basis

$$B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

for $M_{2\times2}$.

[15]

3. Let

$$A = \begin{pmatrix} -6 & 0 & 6 \\ 0 & -3 & 0 \\ -3 & 0 & 3 \end{pmatrix}.$$

(a) Write A as a product of an elementary matrix and an upper triangular matrix.

(b) Use your answer to (a) to find $\det(A)$. Show how you arrived at your answer.

(c) Can A be written as a product of elementary matrices? Give a reason for your answer.

(d) Find a basis for (i) the nullspace of A (ii) the column space of A.

(e) Give the rank and nullity of A.

(f) Find all the eigenvalues of A.

(g) Is there an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$? If so, say why and find P and D; if not, say why not.

[22]

4. Read the proof given below, and then answer the questions following it.

Proof: Let V and W be vector spaces, $T : V \to W$ a linear transformation.
Suppose $\dim(V) = n$ and $0 < \dim(\ker(T)) < n$.
Let $A = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ be a basis for $\ker(T)$. (1)
Choose vectors $\vec{v}_{k+1}, \ldots, \vec{v}_n$ in V such that $B = \{\vec{v}_1, \ldots \vec{v}_k, \vec{v}_{k+1}, \ldots, \vec{v}_n\}$ is a basis for V. (2)
Put $C = \{T(\vec{v}_{k+1}), \ldots, T(\vec{v}_n)\}$.
Now suppose that $a_{k+1}T(\vec{v}_{k+1}) + \cdots + a_nT(\vec{v}_n) = \vec{0}$. (3)
Then it follows that $\vec{u} = a_{k+1}\vec{v}_{k+1} + \cdots + a_n\vec{v}_n \in \ker(T)$. (4)
Hence there are scalars $a_1, \ldots, a_k$ such that $\vec{u} = a_1\vec{v}_1 + \cdots + a_k\vec{v}_k$. (5)
It follows from (4), (5) and the fact that B is a basis that
$a_{k+1} = \cdots = a_n = 0$, (6)
and so C is linearly independent. (7) ■

(a) Why can vectors $\vec{v}_{k+1}, \ldots, \vec{v}_n$ be chosen as at (2)?

(b) How does (5) follow from (4)?

(c) Explain how the statement at (6) is obtained.

(d) Why can we conclude at (7) that C is linearly independent?

(e) Where in the proof is the fact that T is linear used, and how is it used?

(f) What still needs to be proved before we can claim that C is a basis for $\mathrm{ran}(T)$?

(g) If we do know that C is a basis for $\mathrm{ran}(T)$, what can we say about $\dim(\mathrm{ran}(T))$?

(h) Can the argument above still be used if $\dim(\ker(T) = 0$? Give a reason for your answer. If it can be used, state the result proved in this case as concisely as possible. [12]

5. Prove or disprove each of the following statements:

(a) If a vector space V has a linearly independent subset with k elements, then any subset of V with fewer than k elements cannot span V.

(b) Let S_1 and S_2 be finite subsets of a vector space V, then $\text{span}(S_1 \cup S_2) = \text{span}(S_1) \cup \text{span}(S_2)$.

(c) If 0 is an eigenvalue of the square matrix A, then A cannot be written as a product of elementary matrices.

(d) If A is a square matrix such that $A^2 = A$, then A has at most two distinct eigenvalues.

[12]